Atypical Cases and Grading of Epithelial Dysplasia

Atypical cases are semi-quantified using the terms VIN (vulvar intraepithelial neoplasia), PIN (penile intraepithelial neoplasia), or CIN (cervical intraepithelial neoplasia) I, II, or III, depending on the extent of epithelial involvement:
- **CIN I**: Involvement of the lower third of the epithelium
- **CIN II**: Involvement up to two-thirds of the epithelial thickness
- **CIN III**: Involvement of more than two-thirds

When the full thickness of the epithelium is affected, the condition is termed **intraepithelial squamous cell carcinoma (SCC)** or **Bowen's disease (BD)**.

Cockerell et al. highlight the histological and biological similarities between actinic keratosis (AK) and cervical dysplasia, proposing a unified terminology—**KIN (keratinocytic intraepidermal neoplasia)**—for epidermal dysplastic lesions, analogous to CIN/VIN/PIN in mucosal regions.

However, the working group (WG) concluded that unifying histopathological criteria across lesions with different etiologies, pathogenesis, and clinical behaviors offers little practical benefit. Therefore, they recommend retaining the clinically established term **actinic keratosis (AK)** rather than replacing it with a histopathological designation like KIN, which may be less intuitive for clinicians. Histopathologists may still use grading systems (e.g., degree of epithelial dysplasia) in their reports.

AK: Precursor or Squamous Cell Carcinoma?

This classification has implications beyond academic debate, influencing treatment decisions and health insurance policies (e.g., reimbursement, life insurance assessments).

Some authors advocate for a unified concept of SCC, arguing that **AK, Bowen's disease, Bowenoid papulosis, giant condyloma, verrucous carcinoma, keratoacanthoma**, and **proliferating trichilemmal cysts** all represent parts of a spectrum of squamous neoplasia. A key shared feature is the presence of **p53 oncogene mutations**.

Nonetheless, epidemiological data show that AK follows one of three clinical courses:
- Spontaneous regression
- Persistent stability
- Progression to invasive SCC

The risk of progression is low. In the well-cited Australian cohort study by Marks et al. (referenced by NHMRC), the annual progression rate of AK to SCC was less than **1 in 1,000** over a 5-year period.

Current guidelines reflect differing perspectives:
- The **NCCN** classifies AK as **photo-induced precancerous lesions**
- The **NHMRC** and **BAD** do not formally classify AK as cancer but acknowledge that while most cutaneous SCCs arise from AK, the progression risk is minimal. This supports the use of **non-invasive topical therapies** for the majority of AK cases.

Field Cancerization

Originally described in oral carcinomas, the concept of a **cancerization field** refers to an area of skin or mucosa with widespread subclinical genetic damage, predisposing to multiple primary tumors and local recurrences.

In dermatology, chronic **UV radiation** is a major driver of carcinogenesis, contributing to initiation, promotion, and progression. Molecular studies show higher frequencies of oncogenic mutations in sun-exposed skin compared to protected areas. Additionally, genetic abnormalities are often found at the margins of excised AKs.

Clinically, this manifests as:
- Coexistence of SCC and multiple, sometimes subclinical, AKs
- Areas of subtle epidermal roughness detectable only by palpation
- Development of numerous, occasionally confluent AKs—especially in chronically sun-damaged areas such as the scalp

These observations support the idea of treating the **entire affected field** rather than targeting individual lesions. However, **no clinical trials have yet proven the superiority** of field-directed approaches over lesion-specific treatments.

Bowen's Disease (II.2.2)

Bowen's disease is defined as **intraepithelial squamous cell carcinoma**. Its prevalence and incidence in France and globally remain poorly documented.

Epidemiology

- Most commonly diagnosed in the **7th decade of life**
- Affects **women in approximately 70% of reported cases**

Clinical Features

- Presents as a **well-demarcated, discoid, erythematous-squamous plaque**
- Often **keratotic or crusted**
- Typically located on **sun-protected skin**

Lesions progress slowly. In mucosal and semi-mucosal sites (e.g., genitalia), BD may appear:
- **Smooth or weeping**
- **Erythroplastic** (e.g., **erythroplasia of Queyrat** on the penis)
- **Leukoplastic**

Genital forms are frequently associated with **HPV infection**.